Momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate and compare signed momentum.

SUCCESS CRITERIA - I can...

1. Multiply mass by velocity with correct units.
2. Include direction or a sign.
3. Distinguish zero total momentum from no motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “velocity” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

velocity: rate of position change, with direction

mass: a measure of how strongly an object resists acceleration

01 Warm-up: bring back an earlier idea

An asteroid’s period is **5 years**.
1. **Explain why** that number alone cannot establish an Earth encounter.
2. **Name two additional observations needed**.

Word help: encounter = objects reaching nearby positions at the same time

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $p_{total} = p_1 + p_2$.
Use signed velocities; momentum units are kg·m/s.

Worked reasoning: A **3 kg** cart moving left at **2 m/s** has momentum **-6 kg·m/s** when right is positive. The minus sign gives direction, not a negative mass.

Your turn: Draw separate velocity and momentum arrows, labeling their different units.

Sentence starter: Both … . They differ in … because … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The two carts have the same speed in opposite directions. Which momentum values are positive if right is positive?

Sentence starter: I notice … . This supports … because … .



2 kg; 3 m/s right

2 kg; 3 m/s left

Marzano check - circle: ① ② ③

04 Apply the physics

Use the two carts in question **3**: each has mass **2 kg**, and their velocities are **+3** and **-3 m/s**. **Complete the momentum table** and **find the total**. **Explain why a zero total** does not mean either cart is stopped.

Sentence starter: I think … . I can point to …, which shows …, so … .

Object / trial	Mass	Velocity	Momentum
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

One cart doubles mass at the same velocity. Another doubles speed at the same mass. **Compare the momentum multipliers**.

Sentence starter: Both … . They differ in … because … .

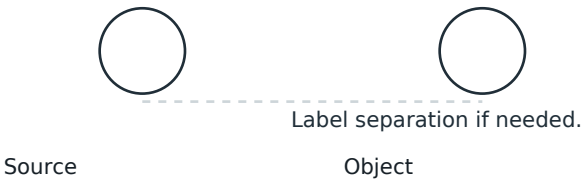
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A **4 kg** test mass feels **24 N** toward a source. **Find g** and the force on a **8 kg** mass at the same point.
Label the masses, field or force relevant to the question.

Word help: field = a description of an influence at each location

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

07 Exit check

A **4 kg** cart moves left at **2 m/s**. **State its momentum** with direction.

Sentence starter: I notice … . This supports … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Build two moving-cart examples whose total momentum is zero but whose masses differ. **Draw both momentum arrows**.

Sentence starter: Both … . They differ in … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③